

Tuninter flight 1153

Flight model: ATR-72 (Turbo prop)
Date: August 06, 2005

From: Italy
To: Tunisian island

Problem

Just couple of miles before the destination, the right engine stops working. They request for an emergency descend to a lower altitude as with one engine it is better to fly in a lower altitude with the air more thicker. Then within couple of minutes the left engine also stops working. The pilots comes to this conclusion from the low fuel pressure warning lights. Knowing that it is not possible to land in the nearest airport, the pilots plans to ditch the plane in the sea. Few passengers had inflated the life vest which can be dangerous and may lead to drowning with less possibility of escape, if water enters the cabin. Without engines running many vital gauges are dead. The pilots instructs ATC to dispatch all rescuers and also tries to land the plane where there is some boat movements so rescue operations can begin early. The pilots must also avoid hitting against any wave as it may break the plane into pieces. Despite the the plane broke into pieces. One survivors report to have lost their life jacket after the impact even though they were not inflated.

Investigation

Pilot survived the crash. Lack of fuel was rules out but there was a low feed pressure warning is crucial clue. The airport is checked for any possible fuel contamination due to sand, sticky residue from refueling trucks and other biological contaminants etc which, turned out to be negative sighting no contamination found. The plane's fuel system seems to have worked. properly. Maintenance log reports a problem with the fuel quality indicator made by the flight before is seems that the **fuel gauge/Fuel quantity indicator (FQI)** installed was not the right gauge. With the wrong FQI, It was confirmed the plane lacked fuel. It was found that the plane could had safely glided to the nearest airport by "**Feathering the propellers**", which means changing the angles of the propellers thereby reducing the drag induced by them. The plane was a little more than required speed for a glide. The more the speed, the more the drag is induced thus reducing the possible gliding distance. But, the crew didn't know the actual cause of the engine failure and was more concern on restarting the engines.

Changes that were brought after the incident

The incident stress on training on rare situation such as twin engine failure. The investigators also recommends airlines to train their pilots on ditching the plane (land on water) when their is twin engine failure. It also stressed ATR to re-design fuel indicators to avoid confusions of installing incorrect part.

Reference

1. The Fatal Error That Led to a Dramatic Ocean Ditching — Mayday: Air Disaster
https://www.youtube.com/watch?v=Ulp_-TDdsvw